



# Python for Fixed Income

COURSE 7 · NLP AND LLMS

## Case Study: Where the Classifier and the Transformer Disagree

WEEK 2 · DAY 5

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Three models, one comparison

## TF-IDF (day 3)

Fitted on 3,650 labelled sections; validation macro F1 0.9271.

## Embedding (day 7)

MiniLM's 384 numbers, then a logistic regression; 0.8882.

## Zero-shot (day 7)

No labelled section, eight descriptions; 0.5951.

Fitted on 2019 to 2023, chosen on 741 sections of 2024; the 735 of 2025 opened once, in 10.6.

# Real documents, described

An 8-K section's label is the filer's item; a model's output is the item it assigns

## Allowed names

- the item the model assigns
- the model's score for an item
- the filer's item

## Never called

- a default signal
- a distress flag or a warning
- a credit event or credit view

**Item 5.02 sections name people: counted by item only; no 5.02 disagreement is read.**

# Where two models part

**274 of 741**

TF-IDF and zero-shot disagree

**51 of 741**

TF-IDF and embedding disagree

**174**

of the first, assigned 1.01 by zero-shot

When TF-IDF and zero-shot part, the filer's item is TF-IDF's 234 times, zero-shot's 9 times, neither's 31 times.

**A disagreement shows where two models had different words to go on.**

# Ten disagreements, read

| #  | Filer | TF-IDF | Zero-shot | Words | TF-IDF's first term |
|----|-------|--------|-----------|-------|---------------------|
| 1  | 1.03  | 1.03   | 1.01      | 300   | dip                 |
| 2  | 2.04  | 2.04   | 1.01      | 300   | payment             |
| 3  | 2.03  | 2.03   | 1.01      | 28    | item 2.03           |
| 4  | 2.04  | 2.04   | 1.01      | 300   | default             |
| 5  | 2.06  | 2.06   | 2.02      | 101   | charge              |
| 6  | 2.03  | 2.03   | 1.01      | 60    | item 1.01           |
| 7  | 1.01  | 2.03   | 1.01      | 154   | the note            |
| 8  | 2.03  | 2.03   | 1.01      | 102   | the note            |
| 9  | 2.04  | 2.04   | 2.03      | 221   | default             |
| 10 | 2.04  | 2.04   | 1.01      | 263   | default             |

## What each model went on

TF-IDF: the terms with the largest parts, its TF-IDF value times the item's coefficient. Zero-shot: the first 120 words, against eight descriptions.

**Disagreements name another item in the body: 0.3540, against 0.1754 of all sections.**

# The case study question

On sections none of the three was chosen on, how often does each assign the filer's item, item by item? Are the differences larger than the noise of which sections were filed?

## **Fix**

the models as chosen

## **Open**

the 735 test sections once

## **Score**

macro F1, recall per item

## **Interval**

paired, 1,000 resamples

**An item is a heading a filer chose; no output here says anything about a company.**

# What each model reads

## TF-IDF

Every word of the stored body, up to 300.

## MiniLM

The first 256 word pieces, then 384 numbers.

## Zero-shot

The first 120 words, against eight sentences.

**0.6544**

of the test sections longer than 120 words

**0.2578**

cut at 300 words by the course's tool

**The descriptions are day 7's, word for word: never reworted after a result.**

# Opened once, checked first

**735**

test sections, 708 filings

**25 of 25**

same zero-shot item live and from the file

**True**

25 live embeddings within 1e-6

```
tolerance = 1e-4 if torch.get_num_threads() == 1 else 1e-3  
# live zero-shot scores equal the committed file's: True; no live score is printed
```

**Filed 2025-01-02 to 2025-12-31: 100 per item, but 1.03 (84) and 2.06 (51).**

# Three models on the 2025 test sections

|             | TF-IDF | Zero-shot | Embedding |
|-------------|--------|-----------|-----------|
| recall 1.03 | 0.8929 | 0.6071    | 0.8214    |
| recall 2.03 | 0.8900 | 0.0900    | 0.8300    |
| recall 2.04 | 0.8900 | 0.2300    | 0.7500    |
| recall 2.06 | 0.9804 | 0.3922    | 0.8039    |
| macro F1    | 0.9366 | 0.6044    | 0.8754    |

The majority item, 1.01 for every section: macro F1 0.0299, accuracy 0.1361.

Each column is the item the model assigns, scored against the filer's item.

# Paired bootstrap intervals

| Difference in macro F1    | Value  | Lower  | Upper  |
|---------------------------|--------|--------|--------|
| TF-IDF minus zero-shot    | 0.3322 | 0.3003 | 0.3664 |
| TF-IDF minus embedding    | 0.0612 | 0.0437 | 0.0822 |
| Embedding minus zero-shot | 0.2710 | 0.2360 | 0.3055 |

```
ensemblekit.bootstrap_interval(truth, positions, paired_difference(a, b), seed=SEED)
```

**Both models scored on the same resampled rows. Intervals that hold 0: 0 of 3.**

# Where it fails

## Sampling

100 per item per year, by the course's design

## One label

147 test sections name another item

## One period

fitted 2019 to 2023, chosen 2024, tested 2025

## The cut

0.2578 cut at 300 words

## The sample

25 recomputed live; the rest trusted

**The zero-shot model assigns 77 of the 100 Item 2.03 sections to 1.01.**

# Excel side by side

## Excel side by side (tested in Excel)

`=COUNTIFS(item range,$A2,assigned range,B$1)`

`=AVERAGE(L2:L9)`

`=COUNTIFS(item range,"2.040")`

`=SUMPRODUCT(--EXACT(item range,"2.040"))`

the confusion matrix: 64 of 64 cells

macro F1: TF-IDF 0.9366

100: the text 2.04 counted

0: exact text only

**COUNTIFS compares criteria that look like numbers as numbers: use EXACT for codes like 2.1 and 2.10.**

# Save the day: the test rows, and week 2

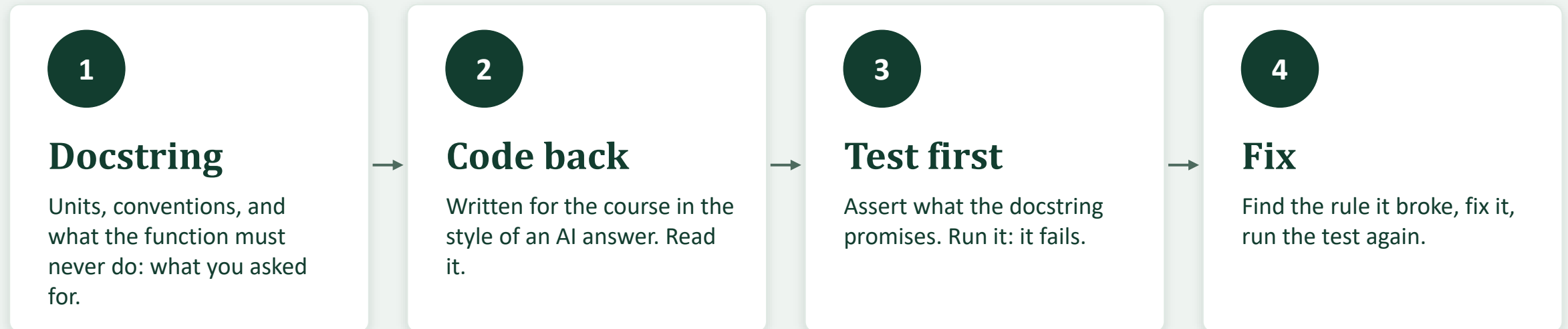
```
git diff results.csv
+tfidf_logistic_balanced,test,0.9366,0.9347,0.8929,0.8900,735
+zero_shot,test,0.6044,0.6299,0.6071,0.2300,735
+embedding_logistic,test,0.8754,0.8748,0.8214,0.7500,735
```

```
git tag week2
git diff --stat week2~1 week2    # results.csv | 3 +++
```

pytest: 51 passed. In my\_bondmath: git diff --stat week1 week2.

# The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: one item against the rest: zero-shot gets 0.8844 on 2.04 (recall 0.2300); never assigning it gets 0.8639.

*Code that runs is not the same as code that is right.*

# What you can do now

- Read where two models part, and why
- Open the test rows once, for chosen models
- Check a cached output with a live sample
- Pair the bootstrap: same rows, both models
- COUNTIFS on codes: use EXACT for look-alikes

**Week 3, Day 1** The first request to an LLM